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Unit 1: Introduction to Databases

What is a DBMS?

A Database Management System (DBMS) is software that manages, stores, retrieves, and manipulates data efficiently.

"A DBMS serves as an interface between the user and the database."

Key Characteristics

- Data Independence --- Separation of data and programs
- Data Integrity --- Accuracy and consistency of data
- Security --- Access control mechanisms
- Concurrency Control --- Multiple users simultaneous access
- Backup & Recovery --- Crash recovery mechanisms

Three-Schema Architecture

1. Physical Level --- How data is stored (blocks, pages)
2. Conceptual Level --- What data is stored (tables, relationships)
3. External Level --- How users view data (views)

Important Terminologies

Term	Definition
Tuple	A single row in a table
Attribute	A column in a table
Domain	Set of allowed values for an attribute
Cardinality	Number of tuples in a relation

Term

Definition

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Basic Operations:

1. Select (σ) --- Filter rows $\sigma_{\text{salary} > 50000}(\text{Employee})$
2. Project (π) --- Select columns $\pi_{\text{name}, \text{salary}}(\text{Employee})$
3. Union ($\cup$) --- Combine two relations
4. Set Difference ($-$) --- Tuples in one but not other
5. Cartesian Product ($\times$) --- Combine all pairs

Additional Operations:

- Join ($\bowtie$) --- Combine related tuples
- Division ($\div$) --- "All" queries

SQL Example

```
SELECT e.name, d.dept_name
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
WHERE e.salary > 50000
ORDER BY e.name;
```

Unit 3: Normalization

Functional Dependencies

Definition: $X \rightarrow Y$ means "X functionally determines Y"

Normal Forms

1. 1NF --- Atomic values only
2. 2NF --- 1NF + No partial dependency on candidate key
3. 3NF --- 2NF + No transitive dependency
4. BCNF --- For every FD $X \rightarrow Y$, X must be a super key

"The key, the whole key, and nothing but the key" ---

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Know ACID properties

Atomicity, Consistency,

Isolation, Durability

- Understand serializability and conflict equivalence
- Remember the difference between primary key and candidate key

End of Notes --- Happy Studying! 🚀